THE NORMAL CLOSURE OF A FINITE SEPARABLE EXTENSION IS GALOIS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Pairs with lemma-separable-first-normal (delivered L48368): together they show normal closures of separable extensions are exactly the Galois closures.

Editorial note. The label lemma-normal-closure-galois names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter definition-normal-closure (source definition): the normal closure of L over K is the minimal normal subextension containing L inside the ambient algebraic closure
- Chapter lemma-separable-first (source lemma): inside a normal algebraic extension M over K sits the tower M over M_sep over K, where M_sep collects the elements separable over K
- Chapter lemma-separable-first-normal (delivered card L48368): M_sep over K is normal
- Chapter convention: Galois means normal and separable

Source reference. Original reference 0BMF: sources/stacks-fields/pinned-original-source.tex, source lines 2341–2341. Prerequisite pointer invoked at 2924; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 030K: sources/stacks-fields/pinned-original-source.tex, source lines 1921–1921. Prerequisite pointer invoked at 2929; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EXK: sources/stacks-fields/pinned-original-source.tex, source lines 2079–2079. Prerequisite pointer invoked at 2930; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be a finite separable extension of fields. Let M be the normal closure of L over K (Definition 0BMF). Then M/K is Galois.*

Proof. The subextension $M/M_{sep}/K$ of Lemma 030K is normal by Lemma 0EXK. Since L/K is separable we have $L \subset M_{sep}$. By minimality $M = M_{sep}$ and the proof is done. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.